

Akash Vijay Nair

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SUMMARY

Final-year BTech Computer Science student with hands-on experience in Python development, backend API development, ETL pipelines, and AI/ML projects. Built and deployed real-world applications involving FastAPI, PostgreSQL, deep learning, and data extraction systems. Strong interest in backend engineering, AI/ML, and software development, with practical experience in building scalable and production-ready applications.

SKILLS

Programming: Python, Java, C, Data Structures & Algorithms, OOP

Database: PostgreSQL, SQLAlchemy

Backend: FastAPI, REST APIs

Machine Learning: Deep Learning, CNN, Ensemble Learning

Tools: Git, Postman, VS Code

PROJECTS

Pneumonia Detection using Ensemble Deep Learning (Web Application)

- Developed an ensemble deep learning model for pneumonia detection from chest X-ray images.
- Combined multiple CNN models to improve classification performance and robustness.
- Built a Flask-based web application for uploading X-ray images and generating real-time predictions.
- Implemented image preprocessing techniques using OpenCV for improved model accuracy.
- Trained on publicly available chest X-ray datasets (e.g., Kaggle).
- Achieved approximately 87% accuracy on validation dataset.
- **Tech Stack:** Python, TensorFlow/Keras, OpenCV, Flask, Deep Learning

Job Application Tracker (Deployed Web Application)

- Built and deployed a full-stack job application tracking system for managing and organizing job applications.
- Implemented CRUD operations, filtering, search, and dashboard analytics.
- Developed backend APIs using FastAPI and integrated PostgreSQL database using SQLAlchemy.
- Built frontend using HTML, CSS, and JavaScript with dynamic API integration.
- Deployed backend, frontend, and database on Render.
- **Tech Stack:** Python, FastAPI, PostgreSQL, SQLAlchemy, HTML, CSS, JavaScript
- **GitHub Repository:** github.com/Aku0921/job-tracker

Automated ETL Pipeline using Python

- Built an automated ETL pipeline to extract data from REST APIs and FTP/SFTP servers.
- Processed and transformed JSON data using Python, storing structured data in PostgreSQL.
- Used tools like Postman for API testing and validation.
- **Tech Stack:** Python, REST APIs, PostgreSQL, JSON, Postman

EDUCATION

SCMS School of Engineering and Technology

Bachelor of Technology in Computer Science Engineering | CGPA: 7.5

2022 – 2026 (Expected)

Kerala, India

Lecole Chempaka

Class XII | Percentage: 75%

2022

Thiruvananthapuram, Kerala

Lecole Chempaka

Class X | Percentage: 92%

2020

Thiruvananthapuram, Kerala

EXPERIENCE

Data Pipeline Intern | UST

May 2025 – June 2025

- Built automated data extraction pipelines using Python to retrieve data from REST APIs and FTP/SFTP servers.
- Processed and transformed JSON data into structured formats for storage in PostgreSQL.
- Integrated external data sources and implemented ETL workflows for efficient data handling.
- Tested and validated API responses using Postman to ensure reliability and correctness.
- Worked on backend data integration and database operations in a real-world development environment.

CERTIFICATIONS

Artificial Intelligence Course HEXINOX

Online

- Completed training in AI fundamentals, including data processing and basic machine learning concepts.

Internet of Things (IoT) Course

Online

- Learned about IoT architecture, sensor integration, and real-world applications of IoT technology.

EXTRA-CURRICULAR ACTIVITIES

Leader KOOTTU (Non-Profit Organization)

- Led initiatives in a student-driven organization supporting people with disabilities.
- Coordinated team activities and conducted events in college.

Cricket Team Member Lecole Chempaka

- Part of U-14 district championship-winning team.
- Developed teamwork, discipline, and competitive skills.